WEB TECHNOLOGY LECTURE - 04

CSS Links

LINKS

- The four links states are:
    - a:link - a normal, unvisited link
    - a:visited - a link the user has visited
    - a:hover - a link when the user mouse over it
    - a:active - a link the moment it is clicked
- When setting the style for several link states, there are some order rules:

```
a:link → a:visited → a:hover → a:active
```

EXAMPLE

```
<html>
<head>
    <style type="text/css">
    a:link {color:red;}    /* unvisited link */
    a:visited {color:green;} /* visited link */
    a:hover {color:blue;}   /* mouse over link */
    a:active {color:yellow;}  /* selected link */
    </style>
</head>
<body>
    <a href="default.asp.html" target="_blank">This is a link</a>
</body>
</html>
```

CSS Lists

CSS LISTS

- Lists come in two basic flavors: unordered and ordered.
- The CSS list properties allow you to:
  - Set different list item markers for ordered lists
  - Set different list item markers for unordered lists
  - Set an image as the list item marker

EXAMPLE

- ul.a {list-style-type:circle;}
- ul.b {list-style-type:square;}
- ol.c {list-style-type:upper-roman;}
- ol.d {list-style-type:lower-alpha;}

VALUES FOR UNORDERED LISTS

Value	Description
None	No marker
Disc	Default. The marker is a filled circle
Circle	The marker is a circle
Square	The marker is a square

VALUES FOR ORDERED LISTS

Value	Description
Armenian	The marker is traditional Armenian numbering
Decimal	The marker is a number
decimal-leading-zero	The marker is a number padded by initial zeros (01, 02, 03, etc.)
Georgian	The marker is traditional Georgian numbering (an, ban, gan, etc.)
lower-alpha	The marker is lower-alpha (a, b, c, d, e, etc.)
lower-greek	The marker is lower-greek (alpha, beta, gamma, etc.)
lower-latin	The marker is lower-latin (a, b, c, d, e, etc.)
lower-roman	The marker is lower-roman (i, ii, iii, iv, v, etc.)
upper-alpha	The marker is upper-alpha (A, B, C, D, E, etc.)
upper-latin	The marker is upper-latin (A, B, C, D, E, etc.)
upper-roman	The marker is upper-roman (I, II, III, IV, V, etc.)

LISTS WITH IMAGES

- To specify an image as the list item marker, use the list-style-image property:

- Code Example

```
ul
{
    list-style-image: url(image.gif');
}
```

CSS Tables

TABLE BORDERS

- The example below specifies a black border for table, th, and td elements:

```
<style>
table,th,td
  {
        border:1px solid black;
  }
</style>
```

COLLAPSE BORDERS

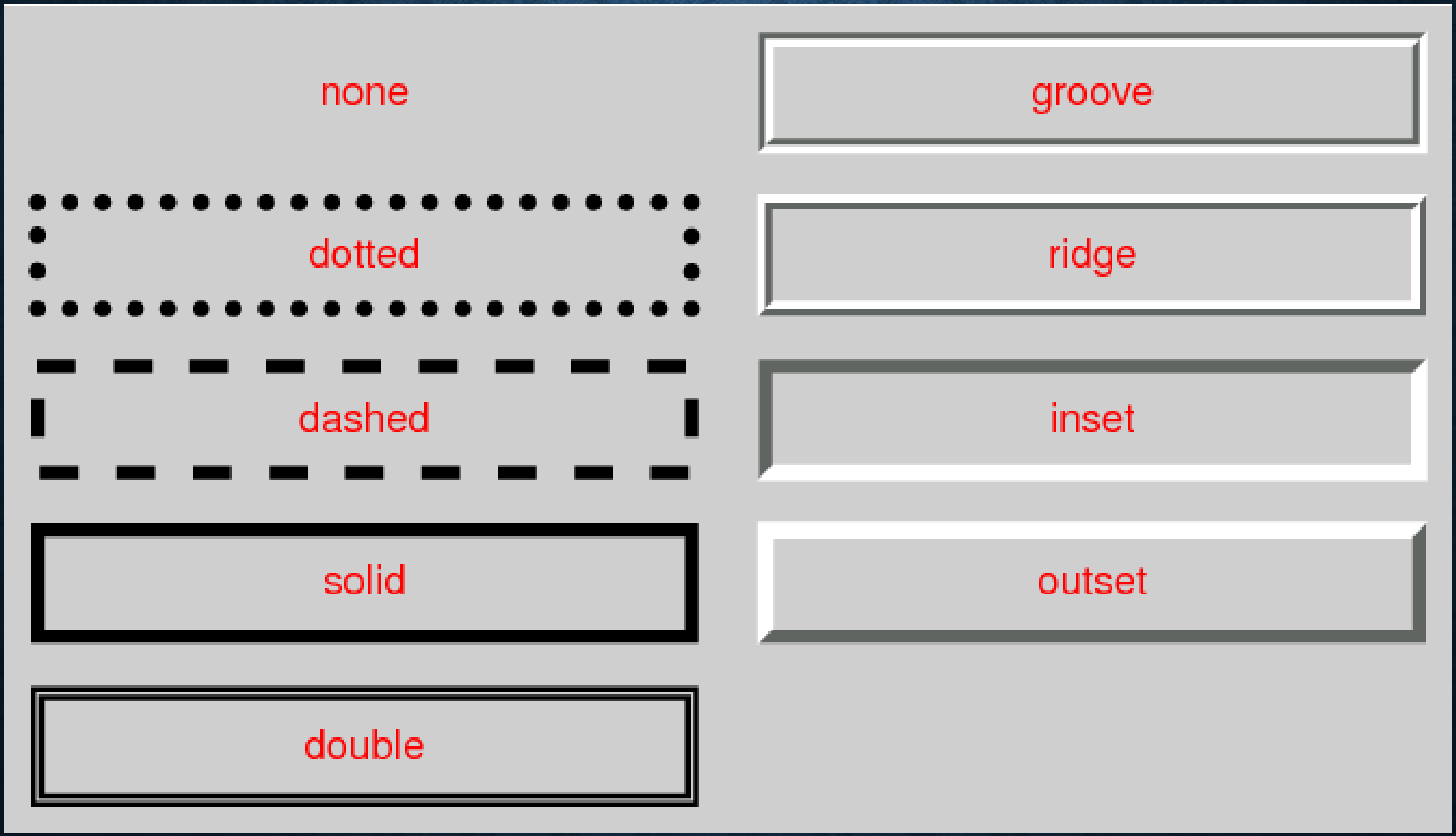
- The table in the example of previous slide has double borders.
- To display a single border for the table, use the border-collapse property.

```
table
{
        border-collapse:collapse;
}
table, td, th
{
        border:1px solid black;
}
```

CSS Border

CSS BORDER

- With the help of CSS every element (not only table) can have borders.



CSS BORDER

- There are numerous types of border styles.
- Example

p.solid {border-style: solid; }

p.double {border-style: double; }

p.groove {border-style: groove; }

p.dotted {border-style: dotted; }

p.dashed {border-style: dashed; }

p.inset {border-style: inset; }

p.outset {border-style: outset; }

p.ridge {border-style: ridge; }

p.hidden {border-style: hidden; }

Border on specific side

border-top-style:dotted;
border-right-style:solid;
border-bottom-style:dotted;
border-left-style:solid;

BORDER WIDTH

- To alter the thickness of your border use the border-width attribute.
- Note: You must define a border-style for the border to show up. Available terms: thin, medium, thick.

```
table { border-width: 7px;}
td { border-width: medium;}
p { border-width: thick; }
```

BORDER COLOR

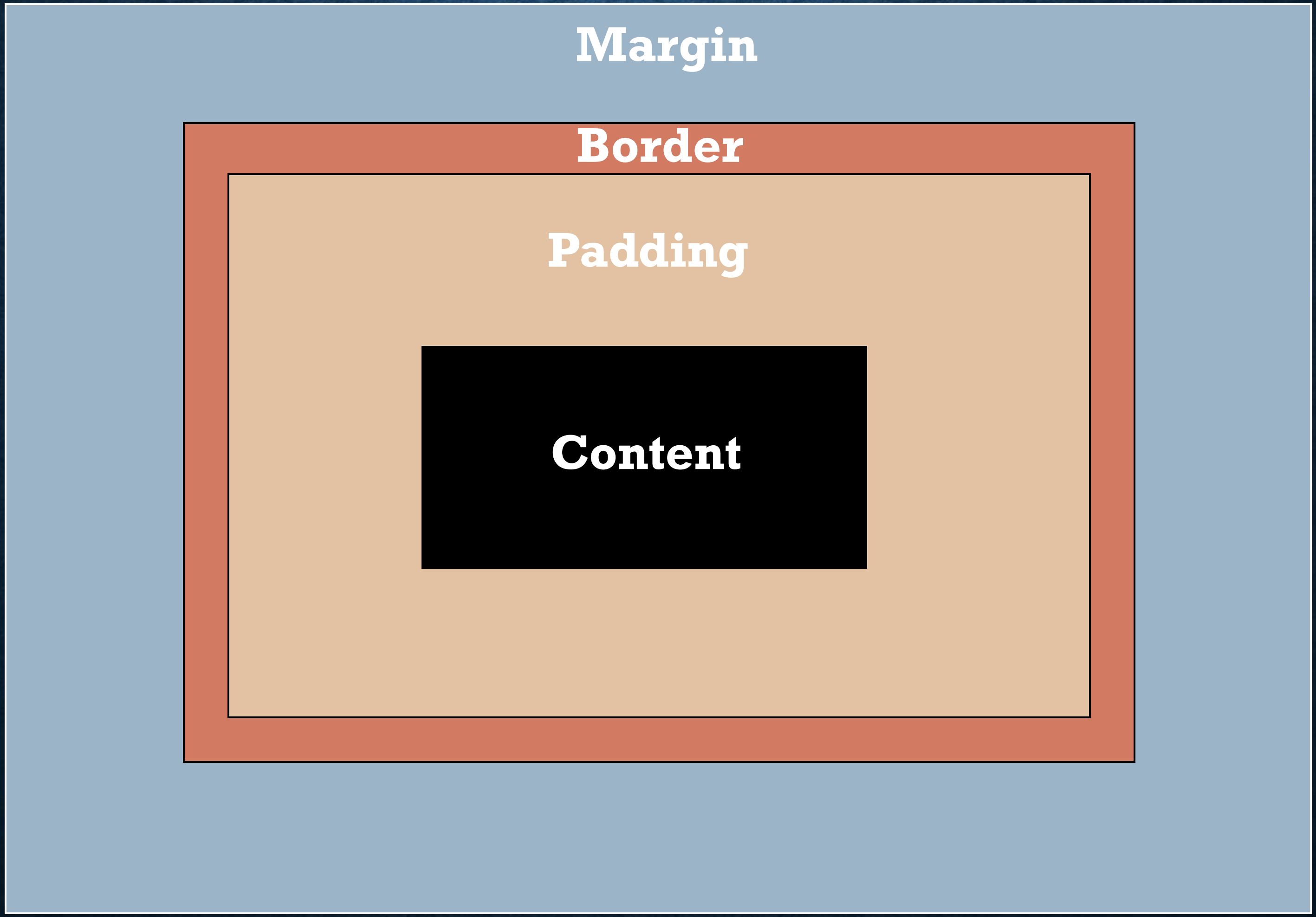
Below is an example of each of these types

```
table { border-color: rgb( 100, 100, 255);}
td { border-color: #FFBD32;
border-style: ridge; }
p { border-color: blue;
border-style: solid; }
```

The CSS Box Model

THE CSS BOX MODEL

Every block element in CSS is effectively inside a box, and can have margins, paddings, borders and contents.



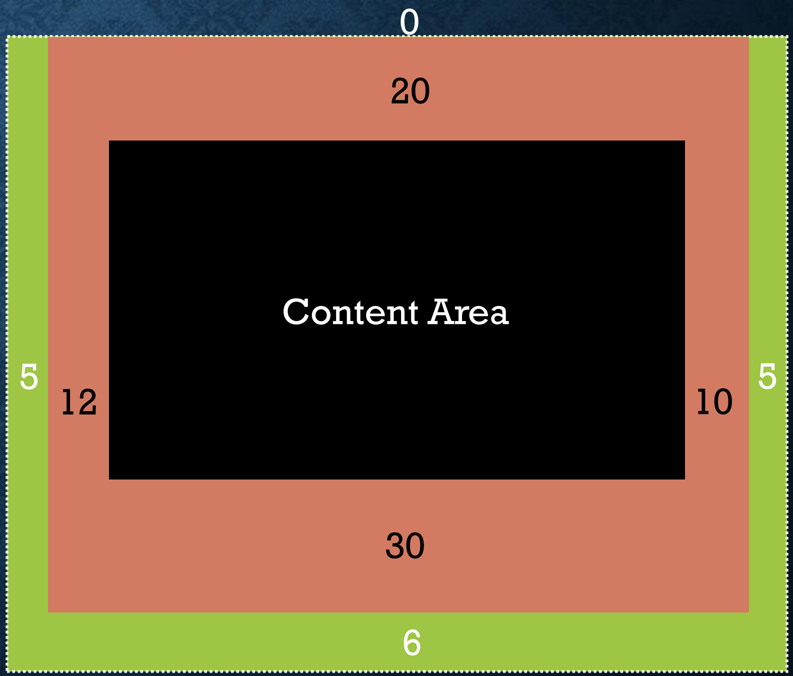
BOX MODEL TERMINOLOGY

- Every CSS box is divided into regions, consisting of:
  1. **Margin** - Clears an area around the border. The margin does not have a background color, and it is completely transparent
  2. **Border** - A border that lies around the padding and content. The border is affected by the background color of the box
  3. **Padding** - Clears an area around the content. The padding is affected by the background color of the box
  4. **Content** - The content of the box, where text and images appear

CSS SHORTHAND: MARGIN & PADDING

- For margin and padding (and others), CSS provides a number of shorthand properties that can save on writing lines and lines of code.
- Instead of writing this:
- #container {

```
        margin-top: 0;
        margin-right: 5px;
        margin-bottom: 6px;
        margin-left: 5px;
        padding-top: 20px;
        padding-right: 10px;
        padding-bottom: 30px;
        padding-left: 12px;
}
```



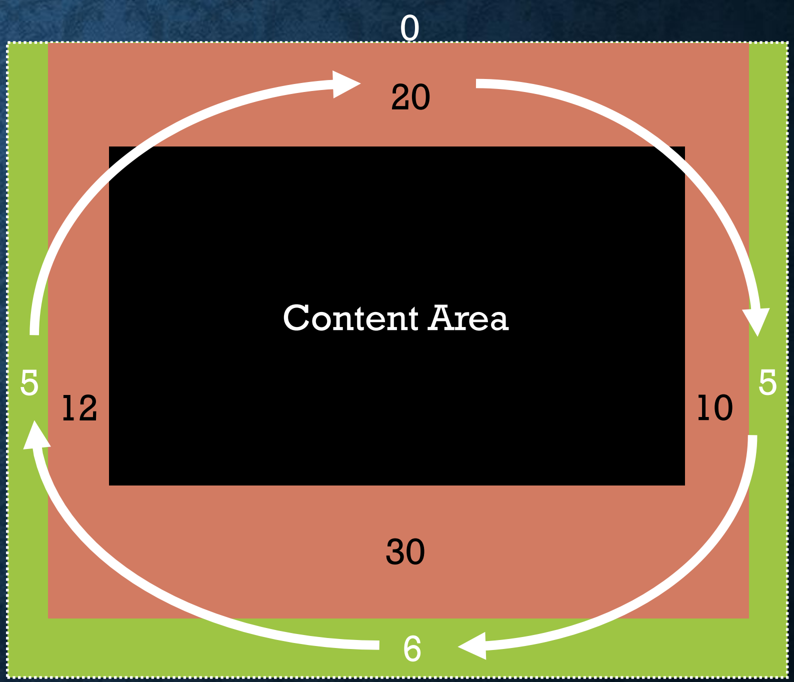
CSS SHORTHAND: MARGIN & PADDING

- …Its much easier to write this:
- #container {

```
#container {
        padding: 20px 10px 30px 12px;
        margin: 0px 5px 6px 5px;
}
```

- The sequence order is always clockwise,

  starting from the top

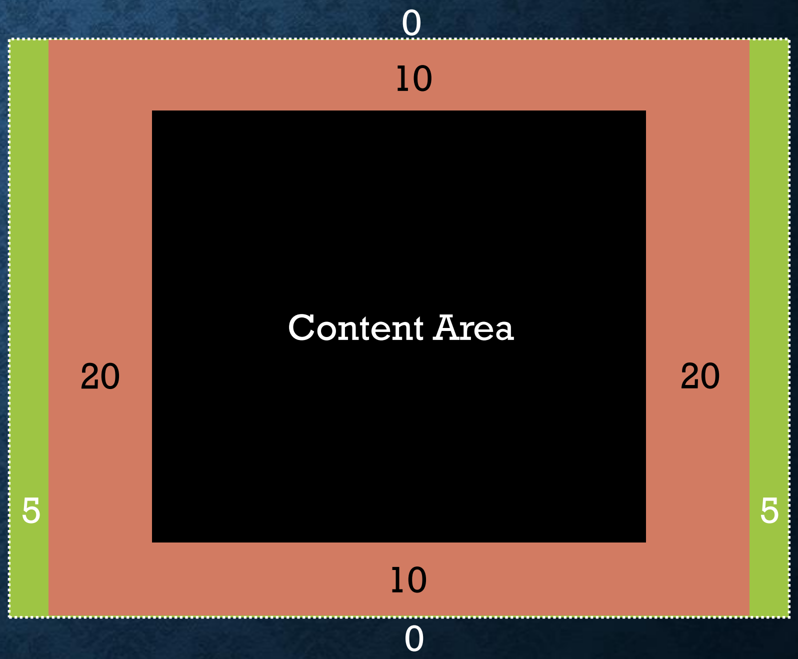


CSS SHORTHAND: MARGIN AND PADDING

- And you can apply two values, example:
- #container {

```
padding: 10px 20px;
margin: 0px 5px;
}
```

- The first value is applied to the top and bottom
- The second value is applied to the left and right



CSS SHORTHAND: MARGIN AND PADDING

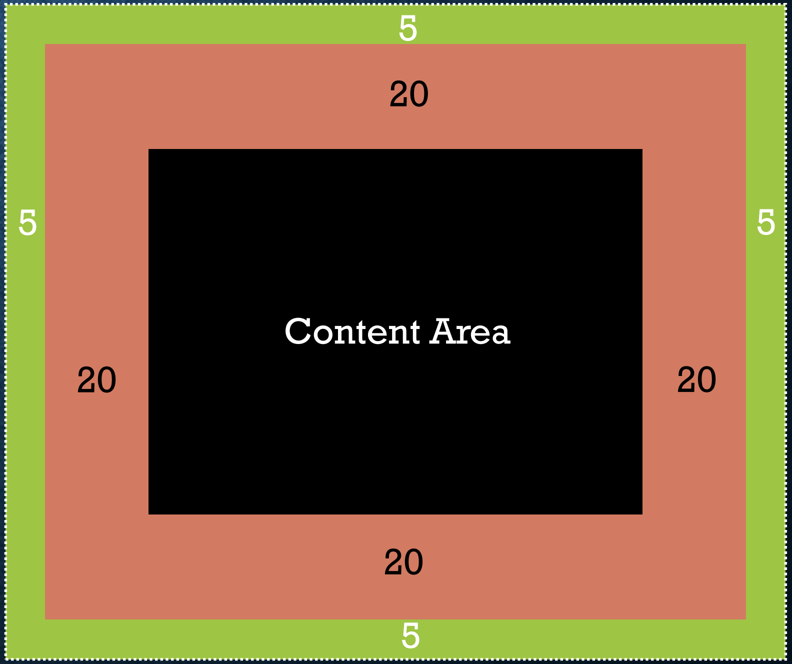
- You can also apply just one value, example:
- #container {

  padding: 20px;

  margin: 5px;

  }

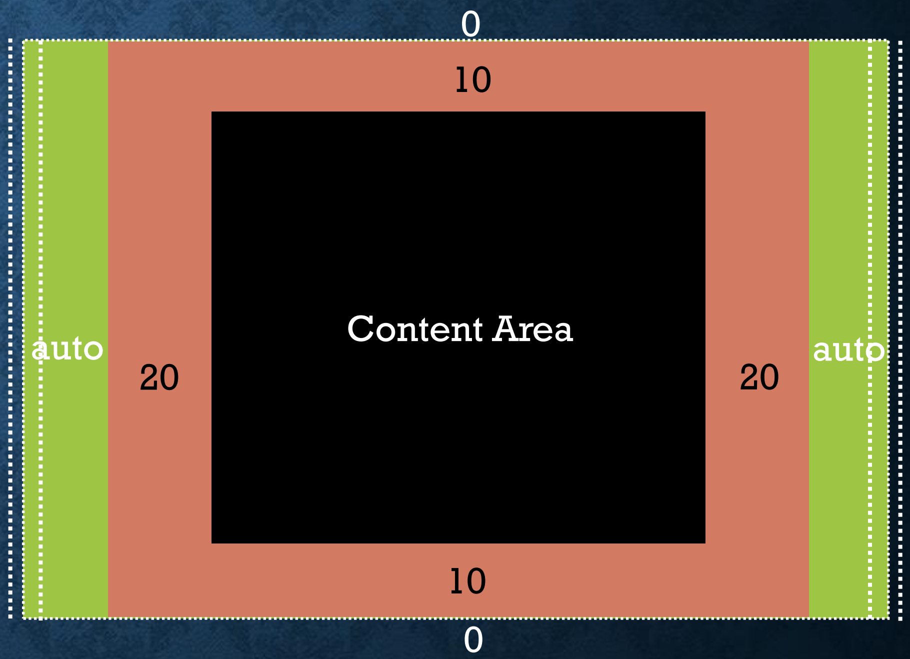
- Which will apply the value specified equally on all 4 sides



CSS SHORTHAND: MARGIN AND PADDING: AUTO

- A useful value to remember is '<u>auto</u>':

```
#container {
        padding: 10px 20px;
        margin: 0px auto;
}
```



- Usually applied to the left & right areas of the margin property, <u>auto</u> is useful for centering a block container element in the browser window

BOX MODEL TYPE

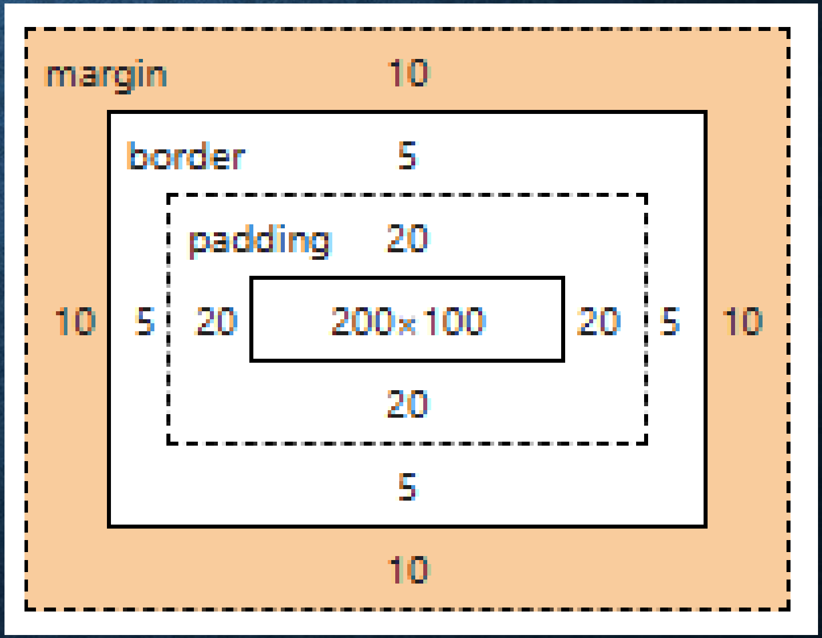
- Box Model Types:
- **Standard Box Model** (default)
- Alternative Box Model:

  CSS: box-sizing: border-box;

STANDARD BOX MODEL

- The width and height properties apply **only to the content**.
- Total size = width + padding + border + margin

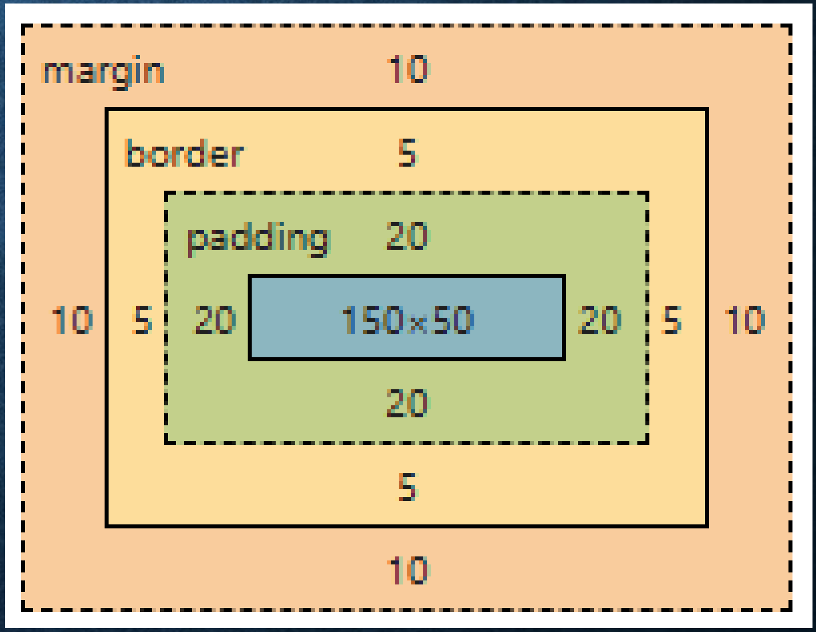
```
.box {
  width: 200px;
  height: 100px;
  padding: 20px;
  border: 5px solid black;
  margin: 10px;
}
```



BORDER-BOX MODEL

- The width and height include content, padding, and border (excluding margin)
- Makes layout calculations easier.

```
.box {
  width: 200px;
  height: 100px;
  padding: 20px;
  border: 5px solid black;
  margin: 10px;
 box-sizing: border-box;
}
```



End